Guillaume **Dalle**

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Researcher in applied mathematics

Professional Experience

Researcher — *2025 - present*
École nationale des ponts et chaussées — *Champs-sur-Marne, France*

- Applied mathematics for transportation systems

Postdoctoral researcher — *2023 - 2024*
École polytechnique fédérale de Lausanne — *Lausanne, Switzerland*

- Theoretical study of graph neural networks

Research intern — *2018*
Électricité de France R&D — *Chatou, France*

- Markovian models for soiling estimation & cleaning optimization in solar power plants

Software development intern — *2017*
Foris AI — *Santiago, Chile*

- Linear Programming & local search for university timetabling

Education

PhD in applied mathematics — *2019 - 2022*
École nationale des ponts et chaussées — *Champs-sur-Marne, France*

- Thesis: *Machine Learning and Combinatorial Optimization Algorithms, with Applications to Railway Planning*

Master of Science in applied mathematics — *2018 - 2019*
École Normale Supérieure Paris-Saclay — *Cachan, France*

- Coursework in machine learning and optimization
- Thesis: *Delay propagation on suburban railway networks*

Engineering degree — *2015 - 2018*
École polytechnique — *Palaiseau, France*

- Coursework in applied mathematics, computer science and humanities

Scientific preparatory classes — *2013 - 2015*
Lycée Louis-le-Grand — *Paris, France*

- Coursework in mathematics and physics

Bilingual French-German high school — *2010 - 2013*
Lycée Jeanne d'Arc — *Clermont-Ferrand, France*

Funded projects

ACME (*Approvisionnement Collaboratif Multimodal Écologique*) — *2025 - 2031*
PEPR MOBIDEC - Agence Nationale de la Recherche

- Multi-disciplinary research consortium to foster mutualization in the supply chain.
- Project website: https://acme-mobidec.github.io/

Publications

[1] L. Bouvier, G. Dalle, A. Parmentier, and T. Vidal, "Solving a Continent-Scale Inventory Routing Problem at Renault," *Transportation Science*, Oct. 2023, doi: 10.1287/trsc.2022.0342.

[2] L. Clarté, A. Vandenbroucque, G. Dalle, B. Loureiro, F. Krzakala, and L. Zdeborova, "Analysis of Bootstrap and Subsampling in High-dimensional Regularized Regression," in *The 40th Conference on Uncertainty in Artificial Intelligence*, June 2024. [Online]. Available: https://openreview.net/forum?id=yZaXk3OxVS

[3] G. Dalle and A. Hill, "A Common Interface for Automatic Differentiation," *Journal of Machine Learning Research*, vol. 27, no. 25, pp. 1–13, 2026, [Online]. Available: http://jmlr.org/papers/v27/25-1024.html

[4] G. Dalle, "HiddenMarkovModels.Jl: Generic, Fast and Reliable State Space Modeling," *Journal of Open Source Software*, vol. 9, no. 96, p. 6436, Apr. 2024, doi: 10.21105/joss.06436.

[5] G. Dalle, L. Baty, L. Bouvier, and A. Parmentier, "Learning with Combinatorial Optimization Layers: A Probabilistic Approach." [Online]. Available: http://arxiv.org/abs/2207.13513

[6] G. Dalle, "Machine Learning and Combinatorial Optimization Algorithms, with Applications to Railway Planning," Doctoral dissertation, 2022. [Online]. Available: https://pastel.hal.science/tel-04053322

[7] G. Dalle and Y. D. Castro, "Minimax Estimation of Partially-Observed Vector Autoregressions," *Electronic Journal of Statistics*, vol. 19, no. 1, pp. 2364–2410, Jan. 2025, doi: 10.1214/25-EJS2387.

[8] G. Dalle and P. Thiran, "Optimal Performance of Graph Convolutional Networks on the Contextual Stochastic Block Model," in *Proceedings of the Third Learning on Graphs Conference*, PMLR, July 2025, pp. 21:1–21:17. [Online]. Available: https://proceedings.mlr.press/v269/dalle25a.html

[9] A. Hill, G. Dalle, and A. Montoison, "An Illustrated Guide to Automatic Sparse Differentiation," in *ICLR Blogposts 2025*, Apr. 2025. [Online]. Available: https://iclr-blogposts.github.io/2025/blog/sparse-autodiff/

[10] A. Hill and G. Dalle, "Sparser, Better, Faster, Stronger: Sparsity Detection for Efficient Automatic Differentiation," *Transactions on Machine Learning Research*, 2025, [Online]. Available: https://openreview.net/forum?id=GtXSN52nIW

[11] A. Montoison, G. Dalle, and A. Gebremedhin, "Revisiting Sparse Matrix Coloring and Bicoloring." [Online]. Available: http://arxiv.org/abs/2505.07308

[12] P. Stephan and G. Dalle, "Method for Determining a Soiling Speed of a Photovoltaic Generation Unit," WO2020115431A1, June 2020 [Online]. Available: https://patents.google.com/patent/WO2020115431A1/en

Talks

Conferences

2025
- RT Optimisation: *Linear and combinatorial optimization on GPUs* (poster session)
- Royal Statistical Society International Conference: *Teaching with Julia* (abstract)
- ROADEF: *Decision-focused learning: software meets theory* (abstract)

2024
- PyData Global: *Automatic differentiation: a tale of two languages* (abstract)
- Julia Optimization Days: *Automatic differentiation: Julia's most confusing superpower?* (abstract)
- JuliaCon: *Gradients for everyone: a quick guide to autodiff in Julia* (abstract)
- JuliaCon: *Fast and generic Hidden Markov Models* (abstract)
- JuliaCon: *Interfaces.jl: base and package interface tests for your objects* (abstract)

2023
- Workshop "Exploring synergies: Machine Learning Meets Physics & Optimization": *Every solution, everywhere, all at once: turning optimization solvers into probability distributions* (abstract)
- Journées Polyèdres et Optimisation Combinatoire: *What is the gradient of a Linear Program? Automatic differentiation on a polytope*
- Julia and Optimization Days: *Graphs in Julia - on the edge of glory* (slides)
- Julia and Optimization Days: *Writing fast Julia code* (slides)

2022
- JuliaCon: *InferOpt.jl: combinatorial optimization in ML pipelines* (recording)
- JuliaCon: *ImplicitDifferentiation.jl: differentiating implicit functions* (recording)
- Journées SMAI-MODE: *Recherche d'itinéraires dans un réseau ferroviaire : apprendre à mieux optimiser* (abstract)
- ROADEF: *Learning to Solve Stochastic Multi-Agent Path Finding* (abstract)

2020
- ROADEF: *Delay propagation on a suburban railway network* (abstract)

2019
- PGMO Days: *Delay propagation on a suburban railway network* (abstract)

Invitations & seminars

2025
- Cycle de conférences SMAI & Musée des arts et métiers: *Des mathématiques pour faire rouler des trains* (web page, slides)
- AdONE Summer School on Optimization-augmented Machine Learning: *Introduction to automatic differentiation* & *Advanced automatic differentiation*
- Laboratoire Jean Kuntzmann, Grenoble: *Sparser, better, faster, stronger - automatic differentiation with a lot of zeros*
- Laboratoire Ville Mobilité Transport: *Mixing data, optimization and decision: the role of automatic differentiation*
- Califrais, Paris: *A tutorial on algorithmic differentiation*
- Julia Meetup Paris: *Automatic differentiation: a tale of two languages*
- Inria Paris: *Sparse automatic differentiation - from theory to practice*

2024
- INDY lab, EPFL: *Sparse automatic differentiation - the fastest Jacobians in the West*
- INDY lab, EPFL: *A primer on algorithmic differentiation*
- Institut de Mathématiques de Bourgogne: *Deep learning meets combinatorial optimization - a tale of missing gradients* (abstract)

2023

- Forum Entreprises & Mathématiques: *Railway to shell: des trains, des maths et du code*
- Swiss Data Science Center: *Every solution, everywhere, all at once: turning optimization solvers into probability distributions*

2022

- MIT CSAIL: *Learning with Combinatorial Optimization Layers: A Probabilistic Approach*

2021

- Institut de Recherche Mathématique de Rennes: *Pourquoi les trains sont-ils toujours en retard?* (abstract)
- École Nationale des Ponts et Chaussées: *Understanding railway delay propagation through latent variable models* (abstract)

Teaching

Teaching assistant & guest lecturer
2025 - present

École nationale des ponts et chaussées — *Champs-sur-Marne, France*

- APOC: Deep learning for combinatorial optimization (course page)
- DECO: Decomposition methods for integer programming (course page))
- MCNDU: Mobilité connectée : nouvelles données, nouveaux usages (course page)
- 1OPTI: Introduction to optimization (course page)

Teaching assistant
2023 - present

École polytechnique fédérale de Lausanne — *Lausanne, Switzerland*

- Statistical physics for optimization & learning (course page)
- Modèles stochastiques pour les communications (course page)

Teaching assistant & guest lecturer
2022 - 2023

Massachusetts Institute of Technology — *Cambridge (MA), United States*

- Julia - solving real-world problems with computation (course page)

Teaching assistant & lecturer
2019 - 2022

École nationale des ponts et chaussées — *Champs-sur-Marne, France*

- Introduction to optimization
- Operations research (course page)
- Design of optimization challenges on:
  - facility location with Air Liquide
  - train shunting with SNCF
  - inventory routing with Renault

Civic service
2015

Lycée Jean-Baptiste Corot — *Savigny-sur-Orge, France*

- Scientific courses and mentoring for struggling high school and undergraduate students

Mentoring

Student advisor
2025 - ongoing

École nationale des ponts et chaussées — *Champs-sur-Marne, France*

- Nicolas Bessin (2026): *Mutualization in the fresh food supply chain*
- Yasmin Van Den Broek (2025): *A Decision-Focused Learning Approach for Multi-Agent Pathfinding*
- Various group projects:
  - optimized dorm room assignment (2022)
  - pathfinding in historical maps (2022)
  - multimodal transit routing (2020)
  - train shunting (2020)

Student advisor
2023 - 2025

École polytechnique fédérale de Lausanne — *Lausanne, Switzerland*

- Antoine Buttier (2025): *Accelerated traffic assignment*
- Antoine Buttier (2024): *Accelerating sparse linear algebra and GraphBLAS with GPUs*
- Chun-Tso Tsai (2023): *Multi-Agent Pathfinding with Mixed-Integer Linear Programming and Lagrange Relaxation*
- Various group and individual projects:
  - parallel graph algorithms (2024)
  - Flatland challenge (2024)
  - graph community detection (2023)

Student advisor
2019 - 2022

École nationale des ponts et chaussées — *Champs-sur-Marne, France*

- Louis Bouvier (2021): *Large Neighborhood Search and Structured Prediction for the Inventory Routing Problem*
- Various group projects:
  - optimized dorm room assignment (2022)

- pathfinding in historical maps (2022)
- train shunting (2020)
- multimodal transit routing (2020)

Community service

Action Editor
ongoing

SCIENTIFIC PUBLISHING

- Action Editor for Transactions on Machine Learning Research

Reviewer
ongoing

SCIENTIFIC PUBLISHING

- Reviewer for scientific journals: Transactions on Machine Learning Research, Journal of Open Source Software, Transportation Science, Transportation Research
- Reviewer for conferences: NeurIPS (2023, 2025), ICML (2023), Learning on Graphs (2024, 2025), JuliaCon (2021-2025)

PhD jury member
2025-ongoing

VARIOUS UNIVERSITIES

- Aurora Rossi (2025): Computational Methods and Analysis of Temporal Networks
- Léo Baty (2025): Combinatorial optimization and decision-focused learning: algorithms, implementations, and applications at Air France

Conference organizer
October 2025

JULIACON LOCAL

Paris

Chair of JuliaCon Local Paris 2025, a european conference on scientific computing and open-source software (170 participants)

Conference organizer
February 2025

ROADEF

Champs-sur-Marne

Co-chair of ROADEF 2025, the national French conference on operations research (>500 participants)

PhD representative
2021

ÉCOLE DES PONTS

Champs-sur-Marne

Representative for PhD students at my university. Advocated for student well-being during Covid lockdowns.

Awards

2025	**Open science prize for research software**, Ministry for higher education and research
2024	**Julia Community Prize**, JuliaCon
2023	**PhD award for mathematics in industry**, AMIES
2021	**Prix Pasquet for an outstanding engineering student**, École des Ponts ParisTech
2015	**Ranked first in the nationwide entrance exam**, École polytechnique

Skills

Languages	English | French | German | Spanish
Programming	Julia | Python | Rust | Git(Hub) | LaTeX | Typst
Personal Interests	Music | Songwriting | Board games | Running | Bouldering